



Image classification based on quantum K-Nearest-Neighbor algorithm

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Abstract

Image classification is an important task in the field of machine learning and image processing. However, common classification method, the K-Nearest-Neighbor algorithm, has high complexity, because its two main processes: similarity computing and searching, are time-consuming. Especially in the era of big data, the problem is prominent when the amount of images to be classified is large. In this paper, we try to use the powerful parallel computing ability of quantum computers to optimize the efficiency of image classification. The scheme is based on quantum K-Nearest-Neighbor algorithm. Firstly, the feature vectors of images are extracted on classical computers. Then, the feature vectors are inputted into a quantum superposition state, which is used to achieve parallel computing of similarity. Next, the quantum minimum search algorithm is used to speed up searching process for similarity. Finally, the image is classified by quantum measurement. The complexity of the quantum algorithm is only $O(\sqrt{kM})$, which is superior to the classical algorithms. Moreover, the measurement step is executed only once to ensure the validity of the scheme. The experimental results show that the classification accuracy is 83.1% on Graz-01 dataset and 78% on Caltech-101 dataset, which is close to existing classical algorithms. Hence, our quantum scheme has a good classification performance while greatly improving the efficiency.

Keywords Quantum K-Nearest-Neighbor · Quantum image classification · Quantum image processing · Machine learning · Quantum intelligence computation

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1 Introduction

In 1982, Feynman proposed a novel computation model, named quantum computation. Due to the superposition and entanglement properties of the quantum states, this novel computation model can efficiently solve some problems that are believed to be intractable on classical computers [1]. After that, many researchers devoted themselves to the research about quantum computation. In particular, in 1994, Shor designed a quantum integer factoring algorithm which can be done in polynomial time and the exponential speedup is achieved compared with the classical algorithm [2]. In 1996, Grover's quantum searching algorithm went quadratically faster than any possible classical algorithms [3]. In addition, quantum image processing (QIP) is an important and rapidly developing sub-domain in quantum computation. Various quantum image representations have been proposed, such as flexible representation of quantum images [4], RGB multi-channel representation [5], novel enhanced quantum representation [6], generalized quantum image representation [7] and Red-Green-Blue multi-channel quantum representation [8]. Some quantum image processing algorithms have been developed based on these representation schemes, such as quantum image scrambling [9,10], quantum image steganography algorithm [11,12], quantum image matching [13], quantum binary images thinning algorithm [14], quantum watermarking [15,16], quantum image edge detection [17], quantum image motion detection [18,19], quantum image searching [20] and quantum image metric [21].

Image classification is an important task in the field of machine learning and image processing, which is widely used in many fields, such as computer vision, network image retrieval and military automation target identification. K-Nearest-Neighbor (KNN) algorithm is one of the typical and simplest methods to do image classification. KNN's basic idea is that if the majority of the k -nearest samples of an image in the feature space belong to a certain category, the image also belongs to this category. It has two core processes: similarity computing and k -nearest samples searching. Since KNN requires no learning and training phases and avoids overfitting of parameters, it has a good accuracy in dealing with classification tasks with more samples and less classes. Researchers propose many improved KNN algorithms [22–30]. However, KNN and its improved algorithms are accompanied by a large amount of computation and have high complexity. In particular, the complexity of similarity computing process is $O(M)$ and the complexity of searching process is $O(M \log k)$, where M is the number of training images.

Recent progress implies that a crossover between machine learning and quantum information processing benefits both fields [31–34]. Quantum mechanics offers tantalizing prospects to enhance machine learning, ranging from reduced computational complexity to improved generalization performance. The most notable examples include quantum enhanced algorithms for principal component analysis [35], quantum support vector machines [36] and quantum Boltzmann machines [37–40]. Ruan proposed a global quantum feature extraction method based on Schmidt decomposition and also proposed a revised quantum learning algorithm that will classify images by computing the Hamming distance of these features [41]. But the features and distances used by this algorithm limit the classification effect. Chen proposed quantum K-Nearest-Neighbor algorithm (QKNN) [42] which realized KNN in quantum

computers. QKNN uses quantum superposition states to achieve parallel computing of similarity and uses the quantum minimum search algorithm to speed up search process for similarity. Compared to classical algorithms, the complexity of similarity computing process is $O(1)$ and the complexity of searching process is $O(\sqrt{kM})$. Reference [43] realized another QKNN algorithm based on the metric of Hamming distance. Its complexity is $O(\sqrt{M}^3)$. Although Refs. [42,43] show the process of QKNN and give the complexity analysis, they do experiments only on the image dataset of ten handwritten digits (0 to 9), instead of natural image datasets.

In this paper, we propose an efficient natural image classification scheme based on QKNN. Firstly, the feature vectors of images are extracted. Then, the feature vectors are stored in quantum superposition state which is used to achieve parallel computing of similarity. Next, the quantum minimum search algorithm is used to speed up the searching process for similarity. Finally, the image is classified by quantum measurement. Moreover, the measurement step is executed only once to ensure the validity of the scheme.

The rest of this paper is organized as follows. Firstly, our scheme is described. Then, complexity analysis and accuracy analysis are given, respectively. Finally, we draw conclusions and outline possible future works.

2 Image classification

In this section, we show how to apply the QKNN algorithm to image classification and describe our solution.

2.1 Basic idea

Image classification is the process that assigns an image to a corresponding class according to certain rules with a high degree of confidence. The basic ideas of our image classification scheme based on QKNN is shown in Fig. 1. Images that have been classified are training images, and the unclassified image is the test image. The test image is classified according to training images. Firstly, the feature vectors of all the images are extracted on a classical computer, which consists of color features and texture features. Then, the feature vectors are inputted into a quantum computer by applying a process for preparing quantum state. Next, distances between the test image and the training images, those are used to describe the similarity between them, are computed in parallel by a quantum circuit and are stored in the amplitude by applying the amplitude estimation algorithm. Then, Dürr's finding minimum algorithm (which will be described in detail in Step 4 in Sect. 2.2) [44] is applied to get the k minimum distances from quantum superposition states [32]. Finally, indexes of the k similar images are obtained by measurement and the final classification result is produced by majority voting.

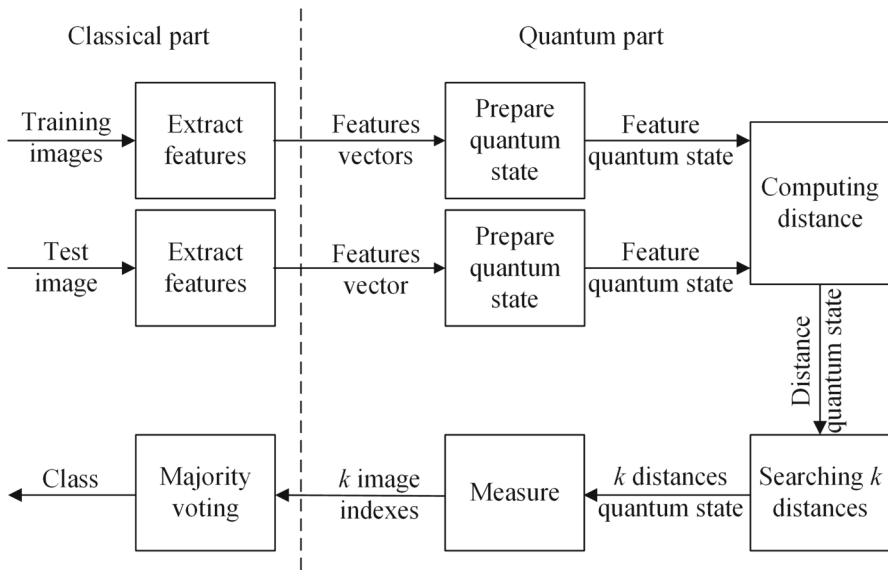


Fig. 1 Algorithm flow chart

2.2 Algorithm

Assume that all the images are RGB (red, green, blue) images. $V = \{V_0, V_1, \dots, V_M\}$ is the complete dataset of the whole task, where M is the number of training images. V_0 is the test image and V_j ($j \in \{1, 2, 3, \dots, M\}$) are training images.

Step 1 Extract features of images.

In this step, feature vectors (including color features and texture features) of the test image and the training images are extracted. The process of feature extraction is divided into three sub-steps.

Step 1.1 Extract color features of images

V_j is transformed from RGB color model to HSB model, where H is the hue, S is the saturation and B is the brightness. Then, according to the human visual characteristics, H space is divided into 8 levels, S space and B space are divided into three levels, respectively, based on Eq. (1), where h , s and b are quantized values of H , S and B .

$$h = \begin{cases} 0, H \in [0, 20] \cup [316, 359] \\ 1, H \in [21, 40] \\ 2, H \in [41, 75] \\ 3, H \in [75, 155] \\ 4, H \in [156, 190] \\ 5, H \in [191, 270] \\ 6, H \in [271, 295] \\ 7, H \in [296, 315] \end{cases}, \quad s = \begin{cases} 0, S \in [0, 0.2) \\ 1, S \in [0.2, 0.7) \\ 2, S \in [0.7, 1] \end{cases}, \quad b = \begin{cases} 0, B \in [0, 0.2) \\ 1, B \in [0.2, 0.7) \\ 2, B \in [0.7, 1] \end{cases} \quad (1)$$

One-dimensional feature vector is synthesized by the 3 components:

$$G = h * Q_h + s * Q_s + b \quad (2)$$

where Q_h and Q_s are weights of h and s , respectively. The larger the value of Q_h and Q_s , the higher the accuracy of quantization, but the more the classification time. In the experiment of Ref. [45], the effect is better when $Q_h = 9$ and $Q_s = 3$. Thus, $G \in \{0, 1, \dots, 71\}$.

Then, by calculating histogram of G , color feature vector $c = (c_1, c_2, \dots, c_{72})$ is obtained. In order to eliminate difference in the size of images, c is normalized.

Step 1.2 Extract texture features of images.

V_j is converted to a grayscale image and the gray level co-occurrence matrix in four directions: $[0, 1]$, $[-1, 1]$, $[-1, 0]$ and $[-1, -1]$, is computed. Then, the mean and the variance of the contrast, correlation, energy and entropy compose the texture feature vector $t = (t_1, t_2, \dots, t_8)$.

Since the physical meaning of each value is different and the size of the value varies, vector t is normalized, which is the texture feature vector.

Step 1.3 Combine texture feature vector and color feature vector.

For the sake of simplicity, the color feature vector c and texture feature vector t are merged into a one-dimensional vector $(c_1, c_2, \dots, c_{72}, t_1, t_2, \dots, t_8)$ which is denoted as $v_j = (v_{j1}, v_{j2}, \dots, v_{ji}, \dots, v_{j80})$. Hence, v_0 is the feature vector of the test image and v_j is the feature vector of the training image V_j ($j \in \{1, 2, \dots, M\}$).

Step 2 Store feature vectors to quantum state.

This step is used to store feature vectors v_0 and v_j ($j \in \{1, 2, \dots, M\}$) into quantum states $|\alpha\rangle$ and $|\beta\rangle$, respectively [46–48].

$$|\alpha\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle \left(\sqrt{1 - v_{0i}^2} |0\rangle + v_{0i} |1\rangle \right) |0\rangle \quad (3)$$

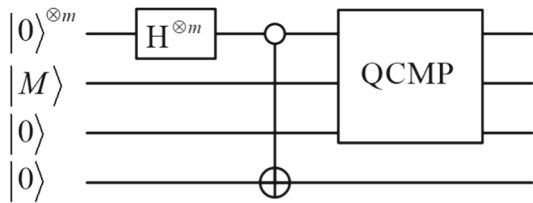
$$|\beta\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle |0\rangle \left(\sqrt{1 - v_{ji}^2} |0\rangle + v_{ji} |1\rangle \right) \quad (4)$$

where $N = 80$ is the dimension of the feature vector.

The preparation consists of two main steps: prepare a number of initial qubits $|0\rangle$ and then store feature vectors into them. Firstly, we describe the preparation of $|\beta\rangle$. From Eqs. (3) and (4), it is obvious that the preparation of $|\alpha\rangle$ is similar to and easier than that of $|\beta\rangle$. The complete preparation process is as follows.

For $|\beta\rangle$, an initial superposition state $|A\rangle = \frac{1}{\sqrt{2^m}} \sum_{j=0}^{2^m-1} |j\rangle$ is prepared by using m Hadamard (H) gates on initial qubits $|0\rangle^{\otimes m}$ as shown in Fig. 2, where $m = \lceil \log_2(M+1) \rceil$. Due to the fact that M may not always be a power of 2, a judgment U_1 is used to choose the $|j\rangle$ s in $|A\rangle$ that satisfy $0 < j \leq M$:

Fig. 2 Quantum circuit of preparing the initial superposition state $\frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle$



$$\begin{aligned}
 U_1(|A\rangle|M\rangle|0\rangle|0\rangle) &= \frac{1}{\sqrt{2^m}} \sum_{j=0} |j\rangle|M\rangle|0\rangle|1\rangle + \frac{1}{\sqrt{2^m}} \sum_{0 < j \leq M} |j\rangle|M\rangle|0\rangle|0\rangle \\
 &+ \frac{1}{\sqrt{2^m}} \sum_{j > M} |j\rangle|M\rangle|1\rangle|0\rangle
 \end{aligned} \quad (5)$$

where $|M\rangle$ is the quantum constant that stores number M as a binary number. U_1 is realized by QCMP (the quantum comparator [49]) model as shown in Fig. 2. QCMP is used to determine j is smaller or bigger than M , and the last two qubits $|0\rangle|0\rangle$ are two flag qubits to store the comparison results. Only if the two flags are in state $|0\rangle|0\rangle$, $|j\rangle$ satisfies $0 < j \leq M$, i.e., j is meaningful. The probability is $M/2^m$. For more details on QCMP, please refer to Ref. [49]. In the following, we only use the subspace that $0 < j \leq M$ and denote it as $\frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle$.

The initial superposition state $\frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle$ is prepared by the same procedure as the preparation of $\frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle$. After the above steps, initial quantum state $|\beta_0\rangle$ is prepared.

$$|\beta_0\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle |0\rangle |0\rangle \quad (6)$$

Next, the feature value is set by using rotation matrix $R_y(2v_{kl})$ and controlled rotation gate U_{kl} .

$$R_y(2v_{kl}) = \begin{pmatrix} \cos v_{kl} & -\sin v_{kl} \\ \sin v_{kl} & \cos v_{kl} \end{pmatrix} \quad (7)$$

$$\begin{aligned}
 U_{kl}|\beta_0\rangle &= \frac{1}{\sqrt{MN}} \left(|0\rangle \otimes \left(\sum_{j=1, j \neq k}^M |j\rangle \langle j| \right) \right. \\
 &\quad \left. \otimes \left(\sum_{i=1, i \neq l}^N |i\rangle \langle i| \right) + R_y(v_{kl}) |0\rangle \otimes |k\rangle \langle l| \right) |0\rangle
 \end{aligned} \quad (8)$$

By defining $U_2 = \prod_{j=1}^M \prod_{i=1}^N U_{ji}$, we can get

$$\begin{aligned}
 U_2 |\beta_0\rangle &= \prod_{j=1}^M \prod_{i=1}^N U_{ji} |\beta_0\rangle \\
 &= \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle \otimes (\cos v_{ji} |0\rangle + \sin v_{ji} |1\rangle) |0\rangle
 \end{aligned} \quad (9)$$

It is denoted as

$$|\beta_1\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle |v_{ji}\rangle |0\rangle \quad (10)$$

and denote the inverse transformation of U_2 as $U^\dagger$.

Then, by acting $R_y(2 \sin^{-1} v_{ji})$ to the last qubit of $|\beta_1\rangle$, we can get

$$|\beta_2\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle |v_{ji}\rangle \left(\sqrt{1 - v_{ji}^2} |0\rangle + v_{ji} |1\rangle \right) \quad (11)$$

This process is denoted as U_3 , where $R_y(2 \sin^{-1} v_{ji})$ is a unitary operation which rotates around the Y axis.

$$R_y(2 \sin^{-1} v_{ji}) = \begin{bmatrix} \sqrt{1 - v_{ji}^2} & -v_{ji} \\ v_{ji} & \sqrt{1 - v_{ji}^2} \end{bmatrix} \quad (12)$$

Finally, apply $U^\dagger$ to $|\beta_2\rangle$ to clear auxiliary qubit $|v_{ji}\rangle$ and we can get $|\beta\rangle$.

The preparation of $|\alpha\rangle$ is similar to that of $|\beta\rangle$. Firstly, an initial superposition state $\frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle$ is prepared using the circuit similar to Fig. 2. Then, apply $\prod_{i=1}^N U_{ji}$, $R_y(2 \sin^{-1} v_{ji})$ and $U^\dagger$ to $\frac{1}{\sqrt{N}} \sum_{i=1}^N |i\rangle |0\rangle |0\rangle$ in turn only by setting $k = 0$ and $j = 0$.

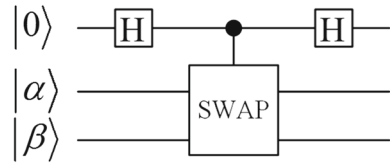
Step 3 Compute distances.

This step computes distances between the test image and the training images by applying controlled swap gate [50,51].

Figure 3 shows the quantum circuit, in which controlled swap gate is used to implement operation: $|\alpha\rangle |\beta\rangle \rightarrow |\beta\rangle |\alpha\rangle$. Firstly, the auxiliary qubit $|0\rangle$ is mapped to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ by a Hadamard gate. Then, this qubit acts as a control qubit for the swap gate to get $\frac{1}{\sqrt{2}}(|0\rangle |\alpha\rangle |\beta\rangle + |1\rangle |\beta\rangle |\alpha\rangle)$. Finally, after the auxiliary qubit passes through another H gate again, quantum state is changed to

$$|\varphi\rangle = \frac{1}{2} |0\rangle (|\alpha\rangle |\beta\rangle + |\beta\rangle |\alpha\rangle) + \frac{1}{2} |1\rangle (|\alpha\rangle |\beta\rangle - |\beta\rangle |\alpha\rangle) \quad (13)$$

The probability of $\varphi = 1$ is denoted as $P(1)$. Thus,

Fig. 3 Controlled swap gate

$$\begin{aligned}
 P(1) &= \langle \varphi | 1 \rangle \langle 1 | \otimes I | \varphi \rangle \\
 &= \left[\frac{1}{2} \langle 0 | (\langle \alpha | \langle \beta | + \langle \beta | \langle \alpha |) + \frac{1}{2} \langle 1 | (\langle \alpha | \langle \beta | - \langle \beta | \langle \alpha |) \right] | 1 \rangle \langle 1 | \\
 &\quad \otimes I \left[\frac{1}{2} | 0 \rangle (\langle \alpha | \langle \beta | + \langle \beta | \langle \alpha |) + \frac{1}{2} | 1 \rangle (\langle \alpha | \langle \beta | - \langle \beta | \langle \alpha |) \right] \\
 &= \left[\frac{1}{2} \langle 1 | (\langle \alpha | \langle \beta | - \langle \beta | \langle \alpha |) \right] \left[\frac{1}{2} \langle 1 | (\langle \alpha | \langle \beta | - \langle \beta | \langle \alpha |) \right] \\
 &= \frac{1}{4} (\langle \alpha | \langle \beta | \langle \alpha | \langle \beta | + \langle \beta | \langle \alpha | \langle \beta | \langle \alpha | - \langle \alpha | \langle \beta | \langle \beta | \langle \alpha | - \langle \beta | \langle \alpha | \langle \alpha | \langle \beta |) \\
 &\hspace{15em} (14)
 \end{aligned}$$

Feature vectors are normalized so that $\langle \alpha | \alpha \rangle = 1$ and $\langle \beta | \beta \rangle = 1$. Hence,

$$\begin{aligned}
 P(1) &= \frac{1}{4} (1 + 1 - \langle \alpha | \langle \beta | \langle \beta | \langle \alpha | - \langle \beta | \langle \alpha | \langle \alpha | \langle \beta |) \\
 &= \frac{1}{4} (2 - 2 |\langle \alpha | \beta \rangle|^2) \\
 &= \frac{1}{2} - \frac{1}{2} |\langle \alpha | \beta \rangle|^2
 \end{aligned} \tag{15}$$

When $|\beta\rangle$ takes a particular state v_j , $P(1) = \frac{1}{2} - \frac{1}{2} |\langle v_0 | v_j \rangle|^2$, where $|\langle v_0 | v_j \rangle|^2$ is the inner product of vector v_0 and v_j . We use $P(1)$ to represent the similarity between vectors and denote it as $d(v_0, v_j)$. Hence, the larger the inner product of the two vectors, the smaller $P(1)$ and the more similar the two images.

Thus, as shown in Fig. 3, by acting controlled swap gate on $|\alpha\rangle$ and $|\beta\rangle$, we can get

$$|\gamma\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle \left[\sqrt{1 - d(v_0, v_j)} |0\rangle + \sqrt{d(v_0, v_j)} |1\rangle \right]. \tag{16}$$

Then, we use amplitude estimation (AE) [52] algorithm to transfer distance information to qubits. Due to the space limitations, please refer to Ref. [52] to get the quantum circuit, the theory proof and more details about AE. This process used R iterations of Grover operators and the error is less than δ , where R and δ satisfy $R \geq \pi(\pi + 1)/\delta$.

AE helps us to get quantum state $|\sigma\rangle$ that stores the similarity information.

$$|\sigma\rangle = \frac{1}{\sqrt{M}} \sum_{j=1}^M |j\rangle |d(v_0, v_j)\rangle \tag{17}$$

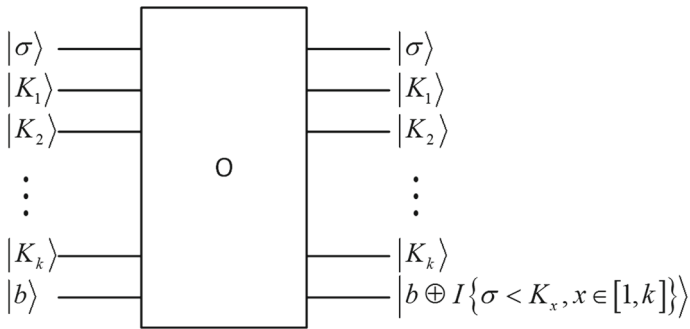


Fig. 4 Circuit model of search process

Step 4 Search for k minimum distances.

In this step, we apply Dürr's algorithm [44,53] to state $|\sigma\rangle$ to find the k minimum distances. Reference [44] shows that this algorithm will return the k minimum after $\sqrt{kM}$ iterations. Figure 4 shows the circuit model of this process. The searching process is as follows.

- (1) Define $K = \{K_1, K_2, \dots, K_k\}$ as k indexes of images that similar to v_0 . Initially, the indexes $K_1, K_2, \dots, K_k \in \{1, 2, \dots, M\}$ are selected randomly.
- (2) By finite iterations of Grover, we can get an index j which satisfies $d(v_0, v_j) < \max\{d(v_0, v_{K_x}), x \in \{1, 2, \dots, k\}\}$. That is to say, the j^{th} training image is more similar to the test image than the image with index K_x .
- (3) Set j to K_{max} , where K_{max} makes $d(v_0, v_{K_{max}}) = \max\{d(v_0, v_{K_x}), x \in \{1, 2, \dots, k\}\}$.
- (4) Repeat (2), (3) and interrupt it after $\sqrt{kM}$ iterations.

For each iteration, the index stored in qubits is replaced by an index with a smaller distance value. Finally, indexes of the k images that are most similar to the test image are stored in auxiliary qubits $|K_1\rangle |K_2\rangle \dots |K_k\rangle$.

Step 5 Measure and classify.

Measure $|K_1\rangle |K_2\rangle \dots |K_k\rangle$ which are the indexes of the first k images that most similar to the test image. According to the basic idea of KNN algorithm, classify the test image to a class whose samples are the majority in results. If there are more than one such classes, choose either of them. In this step, the measurement step is executed only once.

3 Complexity analysis

Our scheme consists of two processes: classical feature extraction and quantum classification based on QKNN, in which the latter is the main part.

In the classical part, the complexity of extracting color features is $O(3M)$, the complexity of extracting texture features is $O(4M)$, and the complexity of combination of features is $O(M)$. Thus, the complexity of Step 1 is $O(8M)$.

We focus on the complexity of the quantum part from the preparation of quantum feature vectors to searching for the k minimum.

Some steps are based on Oracle. In order to discuss the complexity uniformly, we define Oracle as the basic unit. The complexity of each step is analyzed as follows.

Step 2: preparation process of $|\alpha\rangle$ and $|\beta\rangle$ includes U_1 , U_2 , U_3 and $U^\dagger$. U_1 uses some H gates in parallel, one CNOT gate and one quantum comparator. Hence, its complexity is 3. U_2 , U_3 and $U^\dagger$ are regarded as 3 Oracles. This step prepares two quantum states $|\alpha\rangle$ and $|\beta\rangle$, so that it needs 6 Oracles. Hence, the time complexity is $O(1)$, which is a constant.

Step 3: computing distance requires one controlled swap gate, and its complexity is $O(1)$. Reference [52] indicates AE need R iterations of Grover operator, and a Grover operator needs 12 Oracles. Thus, the complexity of this process is $O(12R)$.

Step 4: according to the previous description and the conclusion of Refs. [44,53], the complexity of this step is $O(\sqrt{kM})$.

So the total complexity of the quantum part is

$$O(1) + O(12R) + O(\sqrt{kM}) \approx O(\sqrt{kM}) \quad (18)$$

Established classical image classification algorithm also has the feature extraction step, which is the same as Step 1 of our scheme. That is to say, their complexities are the same. Hence, in the following, we compare the quantum algorithm and the classical algorithm only from the two core processes: similarity computing and searching.

1. If the effect of the dimension of the feature vectors is not taken into account and view a computation distance as a unit, the complexity of the classical similarity computation is $O(M)$. Correspondingly, quantum algorithm only uses a controlled swap gate, whose complexity is a constant $O(1)$. It shows that our algorithm achieves the acceleration from linear complexity to constant complexity.
2. The complexity of searching based on the sorting algorithm is not less than $O(M \log k)$ on classical computers. However, the complexity of quantum search process is $O(\sqrt{kM})$.

We make a detailed comparison between classical search complexity $O(M \log k)$ and quantum search complexity $O(\sqrt{kM})$. Considering the effect of k on the experimental accuracy, $k \in \{3, 5, 7, 9\}$. The result is shown in Fig. 5. Whatever k takes, the complexity of classical search process is higher than the quantum one. Moreover, the larger the M , the greater the quantum advantage. Therefore, quantum search algorithm achieved speedup.

In addition, quantum KNN has two more processes than KNN, which are quantum state preparation and AE. These two processes have only constant complexity. Hence, it is not taken into account.

To sum up, our algorithm significantly reduces the complexity of image classification based on KNN algorithm.

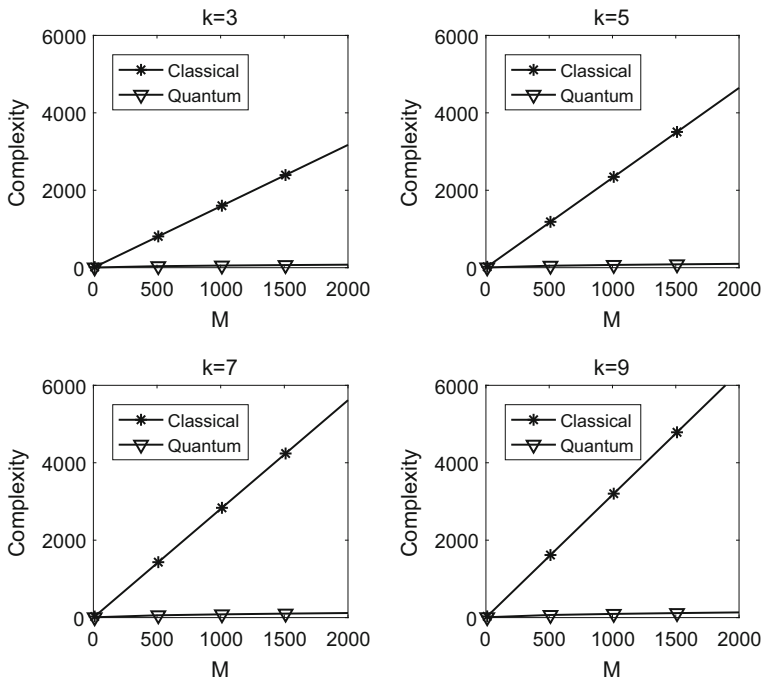


Fig. 5 Complexity comparison between classical and quantum algorithms in search process. M is the number of training images

4 Simulation-based experiments

Section 4.1 gives a simple experiment with 10 images to further show the details of our algorithm. Sections 4.2 and 4.3 use two widely used image sets—Graz-01 and Caltech-101 to demonstrate the accuracy of the algorithm, which have 833 images in 2 classes and 2921 images in 9 classes, respectively.

4.1 A simple experiment

Caltech-101 is a dataset of digital images that is intended to facilitate computer vision research and techniques. Moreover, it is most applicable to techniques involving image recognition classification and categorization [54]. Ten RGB images which include five images from the airplanes class and five images from the Leopards class make up the training dataset as shown in Fig. 6. The test image is shown in Fig. 7.

Firstly, the feature vectors of all the images are extracted according to Step 1. Considering the high dimensionality of the vectors (80), for brevity, only four variables are shown in Table 1.

In Step 2, the feature vectors of the 10 training images are stored in the quantum state $|\beta\rangle$ and the feature vector of the test image is stored in the quantum state $|\alpha\rangle$

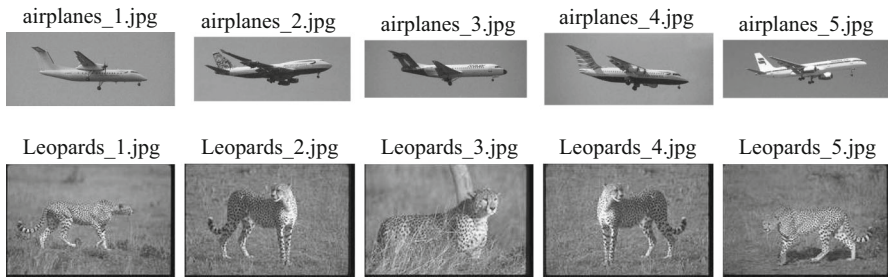


Fig. 6 Training images



Fig. 7 Test image

Table 1 Representative part of the feature vectors

Image	*	v_{*1}	v_{*2}	v_{*24}	v_{*78}
Test image	0	0.00002633	0.00078124	0	0.09610999
airplanes_1	1	0.00008357	0.00005014	0	0.11966605
airplanes_2	2	0.00492092	0.00180055	0	0
airplanes_3	3	0.00196011	0.00013930	0	0.06067670
airplanes_4	4	0.00199697	0.00004851	0	0.09119511
airplanes_5	5	0.00000771	0.00006166	0	0.18334765
Leopards_1	6	0.00085182	0	0.00518836	0.59927777
Leopards_2	7	0.00110141	0	0.01474394	0.57039998
Leopards_3	8	0.00043356	0	0.01250811	0.53275436
Leopards_4	9	0.00127659	0	0.01514383	0.57050397
Leopards_5	10	0.00089379	0	0.01117236	0.56351601

since for each image, its feature vector's dimension is 80 and there are 10 images, i.e., $N = 80$ and $M = 10$,

$$|\alpha\rangle = \frac{1}{\sqrt{80}} \sum_{i=1}^{80} |i\rangle \left(\sqrt{1 - v_{0i}^2} |0\rangle + v_{0i} |1\rangle \right) |0\rangle$$

Table 2 Distances between the test image and the training images

Image	Ranking	$d(v_0, v_*)$
airplanes_1	2	0.0349247032450856
airplanes_2	1	0.0227572679900773
airplanes_3	3	0.0524175219266468
airplanes_4	4	0.0896693757206786
airplanes_5	5	0.126960495310553
Leopards_1	9	0.478449921552929
Leopards_2	8	0.474949680247898
Leopards_3	6	0.470301163393825
Leopards_4	7	0.474904076167589
Leopards_5	10	0.485184863755358

and

$$|\beta\rangle = \frac{1}{\sqrt{10}} \sum_{j=1}^{10} |j\rangle \frac{1}{\sqrt{80}} \sum_{i=1}^{80} |i\rangle |0\rangle \\ \left(\sqrt{1 - v_{ji}^2} |0\rangle + v_{ji} |1\rangle \right).$$

$|\alpha\rangle$ and $|\beta\rangle$ are the inputs of controlled swap gate in Step 3, and we can get distances between the test image and the training images which are stored in $|\gamma\rangle$. In Eq. (19), $d(v_0, v_j)$ is the distance between the test image and the j th training image. Actual data are shown in Table 2.

$$|\gamma\rangle = \frac{1}{\sqrt{10}} \sum_{j=1}^{10} |j\rangle \left[\sqrt{1 - d(v_0, v_j)} |0\rangle + \sqrt{d(v_0, v_j)} |1\rangle \right] \quad (19)$$

Distance values are stored in quantum state $|\sigma\rangle$ by acting AE on $|\gamma\rangle$.

$$|\sigma\rangle = \frac{1}{\sqrt{10}} \sum_{j=1}^{10} |j\rangle |d(v_0, v_j)\rangle \quad (20)$$

In Step 4, by supposing $k=3$, the initial quantum state is $|\sigma\rangle |0000\rangle |0000\rangle |0000\rangle |b\rangle$. After $\lceil \sqrt{30} \rceil = 6$ iterations of Dürr's algorithm, the k smallest distances are searched and are stored in auxiliary qubits. Thus, we can get $|\sigma\rangle |0010\rangle |0001\rangle |0011\rangle |b\rangle$.

Through measuring, results are binary string 0010, 0001 and 0011, i.e., decimal numbers 2, 1 and 3, respectively, which are exactly indexes of the k smallest distances in Table 2. Thus, the training image is similar to the test image airplanes_2.jpgs, airplanes_1.jpgs and airplanes_3.jpgs. Since the three training images are belonged to airplanes class, the test image is also belonged to airplanes class.

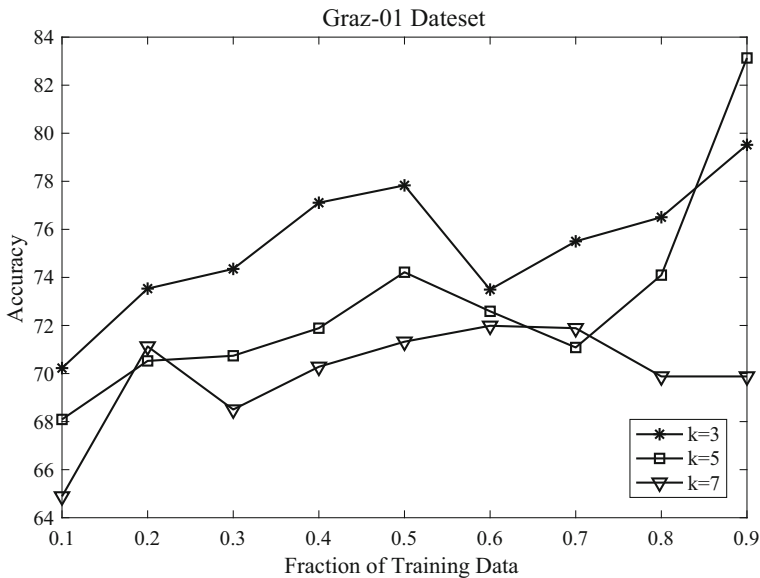


Fig. 8 Performance on the Graz-01 dataset. The meaning of the horizontal axis is to use how many percentage of the 833 images as the training images

Table 3 Comparison of the accuracy of different algorithms

Class	QKNN	Opelt [29]	Lazebnik [30]
Bikes	77	86.5	86.3
Peoples	84.8	80.8	82.3

4.2 Experiment on Graz-01 dataset

The Graz-01 database [55] has two object classes (bikes and persons) and one background class. It is widely used in image classification tasks to compare the accuracy and effectiveness of different methods. We apply our algorithm to this database and demonstrate the performance of our algorithm. Due to characteristics of KNN, we remove the background class of the database. Thus, experiment dataset contains two object classes (bikes and persons) which has total 833 images. Simulation experiment data are shown in Fig. 8.

In the experiments, the maximum value of accuracy is 83.1% when $k = 5$ and using 90% images as training images. When $k = 3$, the average value reaches 75.3% and it is not less than 70% even if the training data are relatively less. It indicates that the algorithm has advantages even if training data are less.

We compare the accuracy of our scheme with the experimental data in Refs. [29,30] because their hypothetical conditions are close. Table 3 shows that the accuracy of Bikes in our algorithm is lower than that of the other two algorithms and the accuracy of People is higher than the others. Overall, the accuracy of our algorithm is almost the same as other algorithms and is acceptable. It shows that our algorithm improves efficiency while ensuring an acceptable accuracy.

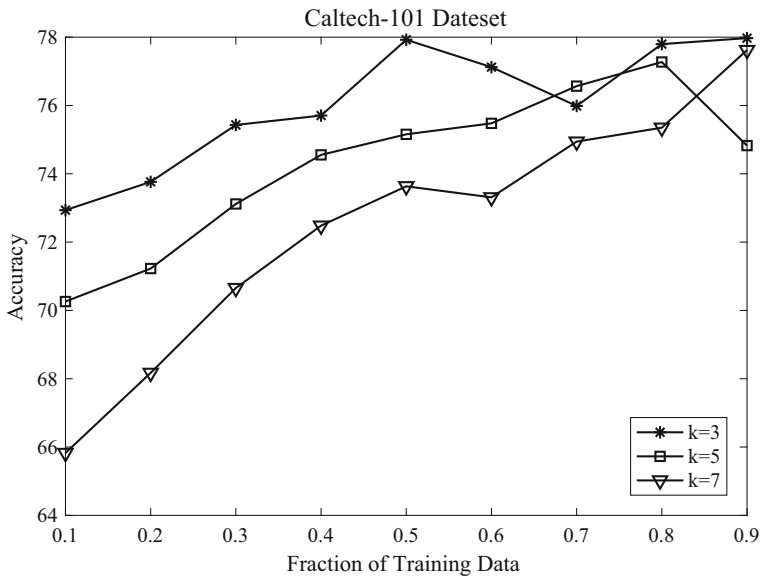


Fig. 9 Performance on the Caltech-101 dataset

4.3 Experiment on Caltech-101 dataset

Caltech-101 has 101 classes (animals, furniture, flowers, etc.) [54]. Since KNN is suitable for classification tasks with more samples and less classes, we choose 9 classes which have more than 100 samples. Hence, our dataset contains 9 classes (airplanes, bonsai, chandelier, etc.) with 2921 images. Figure 9 shows the results, in which the accuracy reaches 78% when $k = 3$ and the training ratio is 90%. Similar to the previous experiment, the accuracy is increased with the increment of training ratio.

Since quantum machine learning is a new research field, and the existing QKNN researches do not do experiments on natural images, there is lack of comparison data in quantum field. Hence, our experimental data are compared only with that of classical algorithms. Through the simulation experiments on Graz-01 and Caltech-101 datasets, the quantum scheme maintains good accuracy while greatly improving the efficiency. Moreover, quantum scheme has an acceptable accuracy even if the training ratio is low.

5 Conclusions

This paper uses the powerful parallel computing ability of quantum computers to optimize the efficiency of image classification. The scheme is based on quantum KNN. Firstly, the feature vectors of images are extracted on classical computers. Then, the feature vectors are inputted into quantum superposition state, which is used to achieve parallel computing of similarity. Next, the quantum minimum search algorithm is

used to speed up the searching process for similarity. Finally, the image is classified by quantum measurement. The complexity of the quantum algorithm is only $O(\sqrt{kM})$, which is superior to the classical algorithms. The measurement step is executed only once to ensure the validity of the scheme. Moreover, our quantum scheme has a good classification performance while greatly improving the efficiency.

There are some things that can be done as future works. Firstly, QKNN is not the only solution for image classification. Image classification algorithms based on more progressive and more subtle machine learning algorithms should be studied to improve the accuracy and application range in the future. Secondly, the method of feature extraction in our scheme is relatively simple and it is done in classical computers. Hence, optimizing feature extraction algorithms and designing quantum implementation solutions is one of the future works.

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